

Why Diabetic Retinopathy Grading?

- Diabetic Retinopathy is a major cause of vision impairment.
- Early detection and correct severity grading are important for timely treatment.
- Manual grading of retinal fundus images requires trained specialists.
- Deep learning can support automated screening and grading.

OUR REVIEW COVERS APPROACHES BASED ON

CNNs and Transfer Learning

Ensemble and Patch-based Models

Self-Supervised Contrastive Learning

Transformers and Multi-View Learning

Graph-Based Learning

Attention Mechanisms

High overall accuracy is not enough if severe and minority DR grades are still misclassified.

Why Five-Class Grading Is Difficult

O

No DR

I

Mild NPDR

2

Moderate NPDR

3

Severe NPDR

4

Proliferative DR

Class Imbalance

No DR and moderate classes often dominate datasets. Severe NPDR and especially PDR have fewer samples.

Adjacent Grades Are Similar

Mild vs Moderate · Moderate vs Severe ·
Severe NPDR vs PDR

Ordinal Nature

No DR → Mild → Moderate → Severe → PDR.
Many models treat them as five completely independent classes.

DR grades follow an order — but conventional models treat them as independent categories.

How DR Deep Learning Methods Have Evolved

○ STAGE 1

CNNs and Transfer Learning

Pretrained CNNs
Image processing + deep learning
Ensemble learning
Patch-based feature extraction

Learn retinal features using strong pretrained visual models.

⦿ STAGE 2

Better Representation Learning

Domain-specific contrastive pretraining
Learning directly from large retinal image collections

Improve the representation before fine-tuning.

● STAGE 3

Explicit Relationship Modeling

Two-field Transformer
Graph-enhanced learning
Attention mechanisms
Quality and uncertainty awareness

Go beyond single-image classification.

MAIN TREND

Single-image backbone → Better representation learning → Explicit modeling of relationships and reliability

Main Ideas from the Key Papers

01

CrossFiT

Fuse two complementary fundus fields from the same eye with Cross-Field Attention and aligned position embeddings, over a ResNet-50 backbone.

84.21% Cohen's Kappa on the reported DRTiD setting, vs 78.06% for the ResNet-50 baseline.

02

FundusNet

Self-supervised contrastive pretraining of a ResNet-50 encoder with Neural Style Transfer augmentation, then external validation.

External UIC test AUC 0.91, vs 0.80 and 0.83 for the reported ImageNet-pretrained baselines.

03

Large EyePACS Study

Five-class grading with a multi-representation CNN ensemble (RGB, HSV, normalized) and explicit minority-class augmentation.

Accuracy improved from 78.06% to 83.78% after augmentation.

04

Graph-Enhanced DL

CNN/Transformer features refined by a Graph Convolutional Network, with image-quality assessment and uncertainty estimation.

Macro-F1, Kappa, AUROC and AUPR — a broader evaluation design than accuracy-only work.

05

Dual Attention

Global Attention and Category Attention Blocks over MobileNetV3-Small, EfficientNet-B0 and DenseNet-169.

Category Attention is motivated by imbalance, but stronger per-grade evidence is still needed.

Comparative Analysis: What Did We Learn?

APPROACH	MAIN CONTRIBUTION	MAJOR STRENGTH	REMAINING LIMITATION
CNN / Transfer Learning	Pretrained visual features	Strong baseline	Dataset dependence
Multi-Representation + Augmentation	Minority data augmentation	Directly addresses imbalance	Accuracy-heavy evaluation
CrossFiT	Two-field feature fusion	Uses complementary retinal views	No dedicated imbalance strategy
FundusNet	Contrastive retinal pretraining	Strong external generalization	Mainly binary classification
Graph-Enhanced DL	Graph + quality + uncertainty	Strong reliability evaluation	Complex pipeline
Dual Attention	Global + category-aware attention	Targets category representation	Minority improvement needs stronger validation

Each approach solves a different part of the problem, but no single approach solves all major challenges.

Research Gaps Identified

GAP 1

Class imbalance remains unresolved

Severe NPDR and PDR remain minority classes. Many studies recognize imbalance; fewer prove strong improvement specifically for rare grades.

GAP 2

Accuracy can be misleading

High accuracy can come from majority classes alone. Evaluation should add Macro-F1, per-class precision, recall and F1, confusion matrix, Kappa/QWK and AUPR.

GAP 3

Ordinal information is underused

Most five-class models use standard categorical classification, although $0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 4$ represents disease progression.

GAP 4 Domain-specific pretraining is not fully explored for five-class grading

FundusNet shows strong contrastive pretraining results, but primarily for binary referral classification.

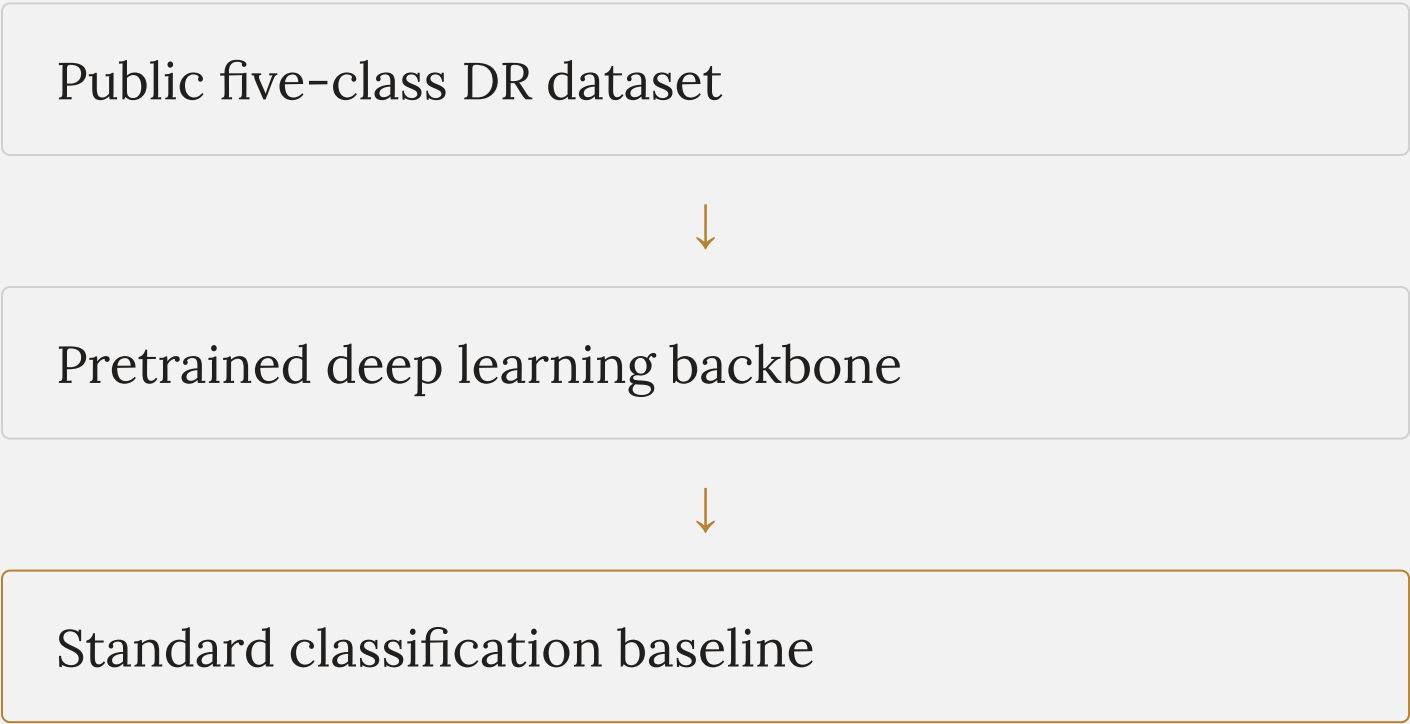
GAP 5 Limited real-world generalization

Performance shifts across hospitals, cameras, image quality, illumination, populations and dataset distributions.

Our Proposed Research Direction

Improve five-class Diabetic Retinopathy grading by addressing class imbalance and the ordinal nature of disease severity.

BASELINE SETUP



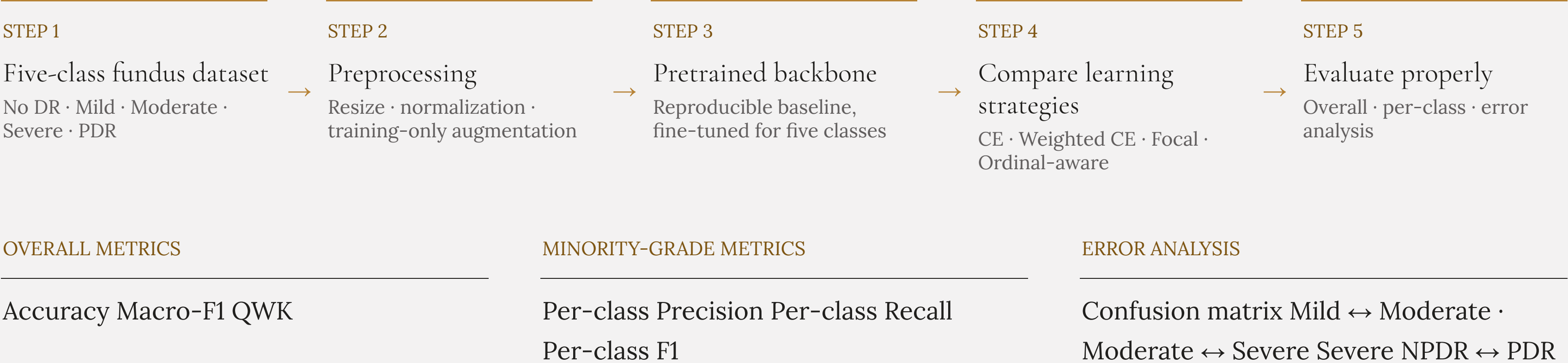
THEN COMPARE TRAINING STRATEGIES

- 01 Standard Cross-Entropy
- 02 Weighted Cross-Entropy
- 03 Focal Loss
- 04 Ordinal-Aware Loss
- 05 Class-Balanced + Ordinal-Aware Approach

RESEARCH QUESTION

Can class-balanced ordinal learning improve Severe NPDR and PDR recognition while reducing clinically large grading errors?

Proposed Methodology and Evaluation



We will not claim improvement based only on accuracy.

Conclusion and Next Steps

WHAT WE LEARNED

- Deep learning has moved from CNN transfer learning toward self-supervised learning, multi-view Transformers, graph-based learning, uncertainty-aware models and attention.
- Transfer learning works, but retinal-domain pretraining is promising.
- Class imbalance remains a major challenge, and minority severe grades need stronger evaluation.
- Five-class grading is significantly harder than binary classification, accuracy alone is insufficient, and the ordinal nature of DR severity is underutilized.
- External validation is essential for clinical applicability.

OUR NEXT STEP

Build a reproducible five-class DR grading baseline and investigate class-balanced ordinal-aware learning through controlled experiments.

FINAL TAKEAWAY

The goal is not only a more accurate model, but a model that better recognizes rare severe disease and makes more clinically meaningful errors.